

Weighted Minimum-Norm and Tikhonov Extensions of the Recursive HSS Solver

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1 Summary

Both extensions reduce *exactly* to the unweighted minimum-norm solver of the companion memo M1 [1]. No new factorization is needed, and the repo’s `H\b` is called unchanged.

- **Weighted minimum norm**, $\min \|Lx\|$ s.t. $Hx = b$, with L diagonal or block diagonal conforming to the leaves. With $y = Lx$, HL^{-1} is HSS on the same tree with only D_τ and V_τ changed (Lemma 3.2). Every step of the solver then either preserves $\|L \cdot\|$ or removes directions that only add to it (Theorem 3.3, Table 1).
- **Tikhonov**, $\min \|S(Hx - b)\|^2 + \lambda^2 \|Lx\|^2$, for *any* shape of H . Augmenting to $[H, \lambda I]$ turns it into a minimum-norm problem (Lemma 4.1). With the identity columns interleaved leaf by leaf, the augmented matrix is HSS on the same tree. It always has full row rank and always satisfies the slack assumption (Lemma 4.2), and every step of the solver preserves Tikhonov optimality (Theorem 4.3, Table 2).
- **Numerics**: accuracy at the conditioning of the weighted or augmented problem, at every λ and for 12 decades of weight range. Cost linear in n up to $n = 2^{19}$, at 0.9–1.25 $\times$ the price of one min-norm solve.
- **Applications** from the list: basis pursuit by IRLS (whose inner loop is exactly the weighted problem) and by Douglas–Rachford (repeated projections), Tikhonov deblurring with λ selection, and k -space-weighted non-Cartesian MRI. Spectral (MWNI) and image-domain weights are not leaf-diagonal and are left open (Section 6.4).

2 Problems and Standing Notation

Notation is that of the companion memo [1] (cited as M1): $H \in \mathbb{C}^{m \times n}$ is HSS on a tree $\mathcal{T}$; a leaf τ has n_τ rows, p_τ columns, generators D_τ, U_τ, V_τ , and the non-leaf generators are R, W, B . Write $\mathcal{M}(H, b) := H^+b$ for the output of the recursive minimum-norm solver, which M1 (Thm. 4.6) proves exact for every full-row-rank HSS matrix with proper ranks (M1, Assumption 2). In this memo L denotes the weight, not the tree depth. We treat

$$(W) \quad x_W = \arg \min_{Hx=b} \|Lx\|_2, \quad H \text{ wide, full row rank,} \quad (1)$$

$$(T) \quad x_\lambda = \arg \min_x \|S(Hx - b)\|_2^2 + \lambda^2 \|Lx\|_2^2, \quad \lambda > 0, H \text{ of any shape,} \quad (2)$$

with weights that are *diagonal or block diagonal conforming to the leaves*:

$$L = \text{diag}(L_\tau)_{\tau \text{ leaf}}, \quad L_\tau \in \mathbb{C}^{p_\tau \times p_\tau} \text{ invertible}; \quad S = \text{diag}(S_\tau), \quad S_\tau \in \mathbb{C}^{n_\tau \times n_\tau}.$$

The diagonal case $L = \text{diag}(w)$, $w_j \neq 0$, is the one used by IRLS and by prior-weighted reconstructions. Note $\|Lx\|^2 = x^* M x$ with $M = L^* L$ SPD, so (W) is the familiar $x_W = M^{-1} H^* (H M^{-1} H^*)^{-1} b$.

Two facts from M1 are used repeatedly.

- (F1) (M1, Lemma 2.1) A leaf's block column is $H(:, J_\tau) = [F_\tau V_\tau^* \text{ (other rows)}; D_\tau \text{ (own rows)}]$, and its block row is $[D_\tau, U_\tau G_\tau]$. Operations on a leaf's columns act on (D_τ, V_τ) only, operations on its rows act on (D_τ, U_τ) only.
- (F2) (M1, Lemmas 4.1–4.4, Table 2) Each operation of the solver either is a unitary change of variables (feasible sets map onto each other, the norm is preserved) or removes variables that enter the norm additively and are free (size reduction) or fixed by the constraints (forced solve).

3 Weighted Minimum Norm

Lemma 3.1 (change of variables). *With $y = Lx$: $x_W = L^{-1}y^*$, where $y^* = (HL^{-1})^+ b$ is the minimum-norm solution of $(HL^{-1})y = b$.*

Proof. Map: $x \mapsto y = Lx$ is a bijection of $\mathbb{C}^n$. Feasible sets: $Hx = b \iff (HL^{-1})y = b$. Objective: $\|Lx\| = \|y\|$. So (W) is the minimum-norm problem for HL^{-1} in the variable y . $\square$

Lemma 3.2 (HL^{-1} is HSS on the same tree). $H_L := HL^{-1}$ has the same tree, ranks, and generators U, R, W, B as H , with leaf generators

$$D_\tau \rightarrow D_\tau L_\tau^{-1}, \quad V_\tau \rightarrow L_\tau^{-*} V_\tau.$$

H_L has full row rank, and it satisfies M1's Assumptions 1 and 2 exactly when H does.

Proof. L^{-1} is block diagonal on the leaf column blocks, so it multiplies each block column $H(:, J_\tau)$ on the right by L_τ^{-1} . By (F1) that block column is $[F_\tau V_\tau^*; D_\tau]$, and $F_\tau V_\tau^* L_\tau^{-1} = F_\tau (L_\tau^{-*} V_\tau)^*$. Nothing else is touched. Right multiplication by an invertible matrix preserves the rank, and $n_\tau, p_\tau, \ell_\tau$ are unchanged. $\square$

Theorem 3.3 (weighted solver). *For (W) with leaf-conforming L , the vector $x = L^{-1} \mathcal{M}(HL^{-1}, b)$ is x_W . In the original variable, every operation of the solver either preserves the weighted norm $\|L \cdot\|$ or removes directions that only add to it, as in Table 1.*

Proof. By Lemma 3.2, H_L is a full-row-rank HSS matrix, so M1 Thm. 4.6 gives $\mathcal{M}(H_L, b) = y^*$, and Lemma 3.1 gives $x_W = L^{-1} y^*$.

For the step-by-step statement, run M1's chain (P0)–(P3) on H_L in the variable y and pull each change of variables back to $x = L^{-1} y$. Since L^{-1} acts leaf by leaf:

- **Size reduction** on H_L : $[V_\tau^* L_\tau^{-1}; D_\tau L_\tau^{-1}] \Omega_\tau = [\tilde{V}_\tau^*, 0; \tilde{D}_\tau, 0]$. In x : $x_\tau = L_\tau^{-1} \Omega_\tau [\tilde{y}_\tau; w_\tau]$, so $\|Lx\|^2 = \sum_\tau \|\tilde{y}_\tau\|^2 + \|w_\tau\|^2$. The discarded w_τ do not affect Hx , so the optimum sets them to zero (M1 Lemma 4.1). These are exactly the directions in which moving x costs weighted norm and buys nothing.
- **Row compression** P_τ^* is unchanged: it acts on rows (on U_τ, D_τ, b), and L acts on columns, so the two commute. It does not change the feasible set.

- **Column decoupling** on H_L : $\tilde{y}_\tau = Q_\tau \hat{y}_\tau$. In x : $x_\tau = L_\tau^{-1} \Omega_\tau [Q_\tau \hat{y}_\tau; 0]$, and $\|Lx\| = \|\hat{y}\|$. This map is an isometry from $(\hat{y}, \|\cdot\|)$ to $(x, \|L \cdot\|)$.
- **Forced solve**: $X^{(1)} = \mathbf{L}^{-1} \hat{b}_1$ is the same for every feasible point, so it contributes the constant $\|X^{(1)}\|^2$ to $\|Lx\|^2$ (M1 Lemma 4.3). Nonsingularity of the triangular blocks follows from full row rank of H_L (M1 Lemma 4.4).
- **Recursion**: the reduced matrix $\bar{H}_L$ is HSS with one level fewer, has full row rank and proper ranks (M1 Lemma 4.4, Sec. 3.4). Induct.

Composing the maps gives $x = L^{-1}y^* = x_W$. □

operation (on $H_L = HL^{-1}$)	map back to x	invariant
weight change	$x = L^{-1}y$	$\ Lx\ = \ y\ , Hx = b \iff H_L y = b$
size reduction	$x_\tau = L_\tau^{-1} \Omega_\tau [\tilde{y}_\tau; w_\tau]$	$\ Lx\ ^2 = \ \tilde{y}\ ^2 + \ w\ ^2 \Rightarrow w = 0$
row compression	$\text{rows} \times P_\tau^*$	feasible set unchanged
column decoupling	$x_\tau = L_\tau^{-1} \Omega_\tau [Q_\tau \hat{y}_\tau; 0]$	$\ Lx\ = \ \hat{y}\ $
forced solve	$\hat{y} = [X^{(1)}; X^{(2)}], X^{(1)} \text{ fixed}$	$\ Lx\ ^2 = c + \ X^{(2)}\ ^2$
recursion / reconstruction	as in M1	M1 Thm. 4.6

Table 1: Weighted version of M1’s Table 2: each step preserves $\|L \cdot\|$ or removes directions that only add to it.

Remark 3.4 (cost and implementation). Forming H_L touches only the leaf generators: $O(\sum_\tau p_\tau(n_\tau + \ell_\tau)) = O(nr)$ work for diagonal L , and $O(\sum_\tau p_\tau^3)$ for dense leaf blocks. The solve costs the same as for H . In MATLAB: `minnorm(H,b,'Weight',w)` or `minnorm(H,b,'Weight',{L_1,...,L_q})`, which forms H_L by editing the leaf generators ($D_\tau L_\tau^{-1}$ and $L_\tau^{-*} V_\tau$, by solves) and calls M1’s solver unchanged. In Python: `hssmn.weighted_minnorm`.

Remark 3.5 (weights that are *not* leaf-conforming). If L couples columns of different leaves, then HL^{-1} is no longer obtained by editing leaf generators. Examples: a circulant or Toeplitz smoothing operator, or a weight that is diagonal in a *different* basis. Two practical cases from the application list are of this type: spectral weights in seismic minimum weighted norm interpolation (MWNI [4]) and image-domain weights in non-Cartesian MRI (Sec. 6). When L is itself HSS, HL^{-1} is HSS of somewhat higher rank, but forming it needs HSS inversion and HSS×HSS products. The repo does not have those yet, and this memo does not treat that case.

4 Tikhonov Regularization

4.1 Standard form via an augmented minimum-norm problem

Lemma 4.1 (augmentation). *For $\lambda > 0$ and any $H \in \mathbb{C}^{m \times n}$, let $[x^*; s^*]$ be the minimum-norm solution of*

$$\begin{bmatrix} H & \lambda I_m \end{bmatrix} \begin{bmatrix} x \\ s \end{bmatrix} = b. \quad (3)$$

Then $x^ = x_\lambda := \arg \min_x \|Hx - b\|^2 + \lambda^2 \|x\|^2$, and $s^* = (b - Hx_\lambda)/\lambda$.*

Proof. Map: for every $x \in \mathbb{C}^n$ there is exactly one s with (x, s) feasible, namely $s = (b - Hx)/\lambda$. So $x \mapsto (x, (b - Hx)/\lambda)$ is a bijection from $\mathbb{C}^n$ onto the feasible set. *Objective:* on the feasible set, $\|[x; s]\|^2 = \|x\|^2 + \lambda^{-2} \|b - Hx\|^2 = \lambda^{-2} (\|Hx - b\|^2 + \lambda^2 \|x\|^2)$. So the two minimizations are the

same problem. Both have unique minimizers: the Tikhonov functional is strictly convex, and (3) has full row rank. $\square$

The closed form agrees: the minimum-norm solution of (3) is $[H^*; \lambda I](HH^* + \lambda^2 I)^{-1}b$, so $x_\lambda = H^*(HH^* + \lambda^2 I)^{-1}b = (H^*H + \lambda^2 I)^{-1}H^*b$. This augmented formulation is classical (it is the “damped” problem of LSQR [3]; see also [6]). The new point is that it *stays HSS on the same tree*, so M1’s solver applies unchanged:

Lemma 4.2 (the augmented matrix is HSS on the same tree). *Order the unknowns leaf by leaf as $(x_{J_\tau}, s_{I_\tau})_\tau$, i.e. apply the column permutation Π that puts the n_τ identity columns belonging to leaf τ ’s rows right after τ ’s own columns. Then $H_a := [H, \lambda I]\Pi$ is HSS on $\mathcal{T}$ with the same U, R, W, B and*

$$D_\tau^a = \begin{bmatrix} D_\tau & \lambda I_{n_\tau} \end{bmatrix} \in \mathbb{C}^{n_\tau \times (p_\tau + n_\tau)}, \quad V_\tau^a = \begin{bmatrix} V_\tau \\ 0_{n_\tau \times \ell_\tau} \end{bmatrix}.$$

Moreover: (i) H_a has full row rank for every $\lambda > 0$ and every shape of H , including square and tall; (ii) Assumption 1 of M1 holds at every leaf as soon as $\ell_\tau \leq p_\tau$ ($\ell_\tau + n_\tau \leq p_\tau + n_\tau$), so even the legacy MATLAB file, which merges levels when the slack is short, never collapses the leaf level of H_a (the current solver never merges); (iii) Π is a permutation, so it maps minimum-norm solutions to minimum-norm solutions.

Proof. The diagonal block of leaf τ in H_a is $[H(I_\tau, J_\tau), \lambda I(I_\tau, I_\tau)] = [D_\tau, \lambda I]$. An off-diagonal block between siblings α, β has columns J_β of H together with the identity columns of I_β . Since $I_\alpha \cap I_\beta = \emptyset$, the identity contributes 0, so the block is $[\mathbf{U}_\alpha B_{\alpha\beta} \mathbf{V}_\beta^*, 0] = \mathbf{U}_\alpha B_{\alpha\beta} [\mathbf{V}_\beta^*, 0]$. With the leafwise interleaving, the zero columns sit under each leaf, which is $V_\tau^a = [V_\tau; 0]$ with the same W above. (i): the columns of λI alone have rank m . (ii): direct. (iii): a permutation is unitary (M1 Lemma 4.2 with $P = I$). $\square$

Theorem 4.3 (Tikhonov solver). *For $\lambda > 0$ and any HSS matrix H with proper ranks, $x_\lambda = [\Pi \mathcal{M}(H_a, b)]_{x\text{-part}}$. In terms of the Tikhonov functional $J_\lambda(x) = \|Hx - b\|^2 + \lambda^2 \|x\|^2$, every operation of the solver applied to H_a preserves J_λ -optimality, as in Table 2.*

Proof. By Lemma 4.2, H_a is HSS with full row rank, so M1 Thm. 4.6 gives $\mathcal{M}(H_a, b) = H_a^+ b$. Undoing Π gives the minimum-norm solution of (3), and Lemma 4.1 identifies its x -part with x_λ .

Step by step, with $z = (x, s)$ and $J_\lambda(x) = \lambda^2 \|z\|^2$ on the feasible set (Lemma 4.1):

- **Size reduction** compresses the $(\ell_\tau + n_\tau) \times (p_\tau + n_\tau)$ stack $\begin{bmatrix} V_\tau^* & 0 \\ D_\tau & \lambda I \end{bmatrix}$. It mixes x_τ and s_τ (data-misfit and solution variables of the same leaf) by a unitary Ω_τ , and always discards $p_\tau - \ell_\tau$ columns, *even when the leaf of H is square or tall*. Along the discarded directions z changes but $H_a z$ does not, so J_λ would only grow: set them to zero.
- **Row compression** acts on U_τ , D_τ^a and b_τ . It is the same P_τ as for H (the identity block becomes λP_τ^* , still part of D_τ^a). It does not change the feasible set.
- **Column decoupling** is a unitary change of z_τ , so it preserves $\|z\|$, hence J_λ .
- **Forced solve**: the forced unknowns are the same for every feasible z . They add a constant to $\|z\|^2$, i.e. to J_λ/λ^2 . Nonsingularity: M1 Lemma 4.4, since H_a has full row rank.
- **Recursion**: the reduced matrix has full row rank and one level fewer. Induct.

Each step is either a unitary change of z or removes components of z that are free or fixed and enter $\|z\|^2$ additively. Composed, the steps map the minimizer of the last problem to the minimizer of J_λ . $\square$

operation	variable / map	invariant
augmentation	$x \mapsto z = (x, (b - Hx)/\lambda)$	$\lambda^2 \ z\ ^2 = J_\lambda(x)$ on $\{H_a z = b\}$
interleave Π	permutation of z	$\ z\ $
size reduction	$z_\tau = \Omega_\tau[\tilde{z}_\tau; w_\tau]$	$\ z\ ^2 = \ \tilde{z}\ ^2 + \ w\ ^2 \Rightarrow w = 0$
row compression	rows $\times P_\tau^*$	feasible set
column decoupling	$\tilde{z}_\tau = Q_\tau \hat{z}_\tau$	$\ z\ $
forced solve	$\hat{z} = [Z^{(1)}; Z^{(2)}], Z^{(1)} \text{ fixed}$	$\ z\ ^2 = c + \ Z^{(2)}\ ^2$
recursion	as in M1	M1 Thm. 4.6

Table 2: The chain behind Theorem 4.3.

4.2 General form, data weights, limits

Proposition 4.4 (general form). *For (T) with leaf-conforming S and invertible leaf-conforming L , put $\tilde{H} = SHL^{-1}$, $\tilde{b} = Sb$. Then $x_\lambda = L^{-1}\tilde{y}_\lambda$, where $\tilde{y}_\lambda$ is the standard-form Tikhonov solution for $(\tilde{H}, \tilde{b})$. $\tilde{H}$ is HSS on the same tree with $D_\tau \rightarrow S_\tau D_\tau L_\tau^{-1}$, $U_\tau \rightarrow S_\tau U_\tau$, $V_\tau \rightarrow L_\tau^{-*} V_\tau$, so Theorem 4.3 applies.*

Proof. With $y = Lx$ the objective is $\|\tilde{H}y - \tilde{b}\|^2 + \lambda^2 \|y\|^2$, a bijective change of variables. The structure follows from (F1) on rows and on columns, as in Lemma 3.2. $\square$

S may be singular: a 0/1 mask that drops corrupted data rows keeps $[\tilde{H}, \lambda I]$ at full row rank. *Limits.* As $\lambda \rightarrow 0$, $x_\lambda \rightarrow H^+b$: the minimum-norm solution for wide full-row-rank H , the least-squares solution for tall full-column-rank H . So the Tikhonov path connects to M1 continuously. As $\lambda \rightarrow \infty$, $x_\lambda \approx \lambda^{-2}H^*b$. *Conditioning.* $\kappa(H_a)^2 = (\sigma_1^2 + \lambda^2)/(\sigma_{\min}^2 + \lambda^2)$ for wide or square H , and $(\sigma_1^2 + \lambda^2)/\lambda^2$ for tall H . For wide and square H , regularization improves the conditioning of the problem the solver actually sees. For tall H , $\kappa(H_a) \rightarrow \infty$ as $\lambda \rightarrow 0$ although x_λ tends to the least-squares solution, whose conditioning is that of H ; there $\kappa(H_a)u$ is a loose bound for the error in x (Section 5.2). *Cost.* Augmented leaves have $p_\tau + n_\tau$ columns, but the size reduction brings them back to $\ell_\tau + n_\tau$, the width a leaf of H has after its own size reduction when its slack suffices. Only that first QR is larger, by at most the factor $1 + n_\tau/p_\tau$; every later operation has the same sizes as for H . So a Tikhonov factorization costs little more than a minimum-norm one (measured: 1.0–1.25 $\times$). Each λ needs a new factorization, since D_τ^a depends on λ . A λ sweep of length K costs K factorizations, each linear in n . In MATLAB: `[x,r] = tikhonov(H,b,lambda,'Weight',L,'DataWeight',S)`, with L and S vectors or cells of leaf blocks. In Python: `hssmn.tikhonov`.

4.3 Algorithms

Algorithm 1 Weighted minimum norm and Tikhonov through $\mathcal{M}$

```

1: function WEIGHTEDMINNORM( $H, b, L$ )                                ▷ minnorm( $H, b, \text{'Weight'}, L$ )
2:    $D_\tau \leftarrow D_\tau L_\tau^{-1}, V_\tau \leftarrow L_\tau^{-*} V_\tau$  for every leaf      ▷  $H_L = HL^{-1}, O(nr)$ 
3:   return  $L^{-1} \mathcal{M}(H_L, b)$ 
4: end function
5: function TIKHONOV( $H, b, \lambda, S, L$ )                                ▷ tikhonov
6:    $D_\tau \leftarrow S_\tau D_\tau L_\tau^{-1}, U_\tau \leftarrow S_\tau U_\tau, V_\tau \leftarrow L_\tau^{-*} V_\tau; \tilde{b} = Sb$       ▷ general  $\rightarrow$  standard form
7:    $D_\tau \leftarrow [D_\tau, \lambda I_{n_\tau}], V_\tau \leftarrow [V_\tau; 0]$  for every leaf      ▷ augment + interleave
8:    $z \leftarrow \mathcal{M}(H_a, \tilde{b}); y \leftarrow z_{x\text{-part}}; r \leftarrow -\lambda z_{s\text{-part}}$       ▷  $r = S(Hx - b)$ 
9:   return  $x = L^{-1}y, r$ 
10: end function

```

5 Numerical Experiments

Protocol, implementations and test families are those of M1 (Sec. 6.1, Table 3). Python reference timings are single-threaded medians after a warm-up (5 runs for $n \leq 2^{16}$, 3 above), on an otherwise idle machine. The current repo solver is cross-checked under Octave in Table 6 of M1, which includes the weighted and Tikhonov wrappers: they agree with the Python reference to $\leq 4.2 \times 10^{-14}$.

5.1 Weighted minimum norm

W1 (accuracy). Diagonal weights $L = \text{diag}(10^{\alpha u_j}), u_j \sim U[-\frac{1}{2}, \frac{1}{2}]$, for $\alpha = 0, 2, \dots, 12$, i.e. up to 12 orders of magnitude between the smallest and largest weight. Leaf-block weights $L_\tau = \text{chol}(I + \beta K_\tau)$ with a Gaussian smoothing kernel K_τ (width 3), $\beta = 1, 10^2, 10^4$. Families F2 (1024×2048), F3 (683×2046) and F4 (1016×2048). The reference is the dense backward-stable solution $L^{-1} \text{QR-min-norm}(HL^{-1}, b)$. Figure 1: the error follows $\kappa(HL^{-1})u$, the conditioning of the weighted problem. It is at or below that line for $\kappa \gtrsim 30$, over 9 decades (κ up to 10^{10}); for better-conditioned cases it sits at a floor of $1\text{--}10 \times 10^{-15}$, up to a few times above κu . The weighted norm of the computed solution matches the reference to 10 digits or better in every case. The block weights give errors of $1\text{--}10 \times 10^{-15}$.

W2 (scale). F2 with $\alpha = 6$ and manufactured weighted solutions $x_\star = L^{-1}(HL^{-1})^* z$ (exact for the HSS matrix), up to $n = 2^{19}$: the error stays at 2.7×10^{-15} and the residual at 6.4×10^{-16} . The cost is that of an unweighted solve, $0.9\text{--}1.13\times$ from $n = 2^{12}$ to 2^{19} (0.50 s against 0.49 s at $n = 2^{16}$), because forming HL^{-1} touches only the leaves.

5.2 Tikhonov

T1 (accuracy along the λ path). $\lambda/\|H\|_2 = 10^{-8}, \dots, 10^2$ for wide (1024×2048), square (1536^2) and tall (2048×1024) F2, for the wide stride-2 “valid” Gaussian convolution (1016×2048 , F4) and for a square stride-1 “same” Gaussian convolution (1536^2). The reference is the dense SVD filter solution $\sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda^2} (u_i^* b) v_i$. Figure 2a: errors of $4 \times 10^{-15}\text{--}4 \times 10^{-13}$ for the F2 matrices and the wide convolution at every λ , including the tall case, where plain min-norm does not apply. For the tall case $\kappa(H_a)u$ is a loose bound: at $\lambda = 10^{-8}\|H\|$ it is 10^{-8} , the error 4×10^{-13} (Sec. 4). For the square convolution ($\kappa = 2.4 \times 10^7$) the error at small λ is 5×10^{-10} , below $\kappa(H_a)u = 2.6 \times 10^{-9}$, and it falls to 10^{-14} as λ grows and $\kappa(H_a)$ drops. At $\lambda = 100\|H\|$ the relative error of x rises to 10^{-13} : there $\|x\| \approx \|H\|\|s\|/\lambda$ is a small part of $z = (x, s)$. The general form (S, L diagonal with 2 decades each) matches a dense stacked least-squares solve to 9×10^{-13} ($\lambda = 10^{-4}$) and 1.2×10^{-14} ($\lambda = 1$).

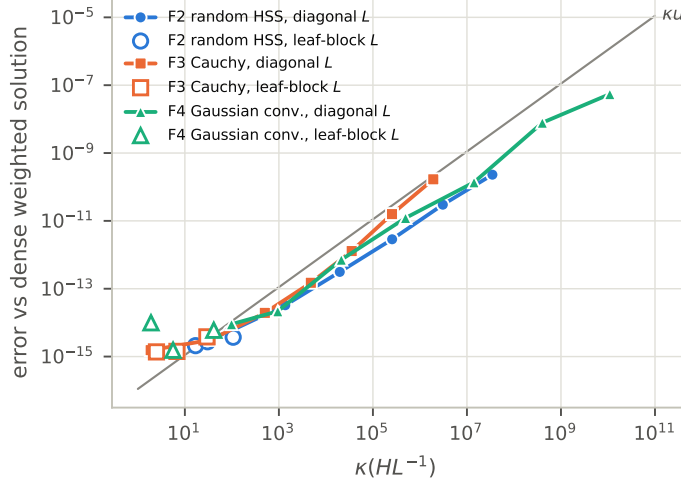


Figure 1: W1: weighted minimum-norm error against $\kappa(HL^{-1})$. Filled: diagonal L with up to 12 decades of dynamic range. Open: leaf-block SPD L . Grey: κu .

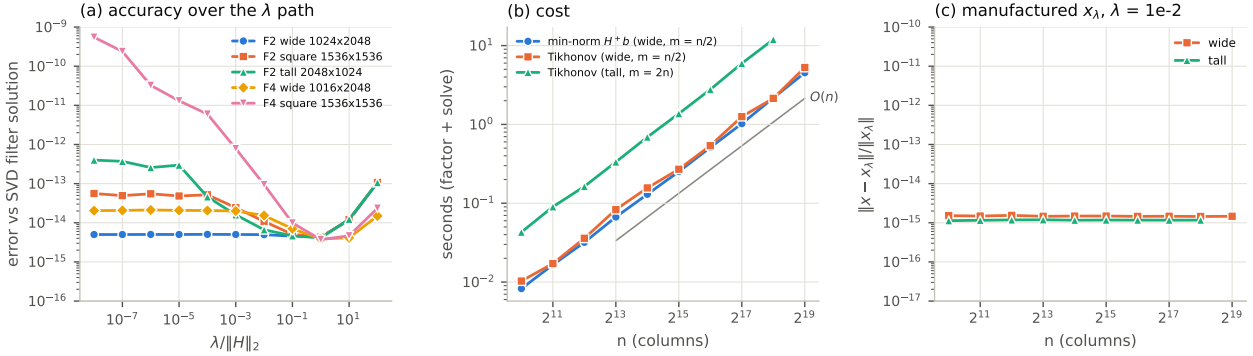


Figure 2: T1–T3. (a) Error against the SVD filter solution along the λ path. (b) Cost against min-norm. (c) Manufactured Tikhonov solutions at scale.

T2/T3 (scale and cost). Manufactured Tikhonov solutions ($x_\star = H^\star z$, $b = Hx_\star + \lambda^2 z$, so $x_\lambda = x_\star$ exactly) with $\lambda = 10^{-2}$: error 1.5×10^{-15} (wide, up to $n = 2^{19}$) and 1.2×10^{-15} (tall $m = 2n$, up to $n = 2^{18}$), flat in n (Figure 2c). The cost is linear (Figure 2b). A Tikhonov solve costs $1.0\text{--}1.25\times$ a min-norm solve of the same wide H , up to $n = 2^{19}$ (5.3s against 4.5s there). That is below the factor $1 + m/n = 1.5$ one might expect from the wider leaves, because the size reduction removes the extra columns at once (Sec. 4).

6 Applications

6.1 Basis pursuit: IRLS and Douglas–Rachford

Problem. Seismic reflectivity is modelled as a sparse spike train convolved with a band-pass wavelet. $x_0 \in \mathbb{R}^{4096}$ has 60 spikes (amplitudes $\pm[0.5, 1.5]$, separation ≥ 24 samples). H is the Ricker wavelet ($\sigma = 2.5$) convolution with stride 2, F4, 2038×4096 , and $b = Hx_0$. We solve basis pursuit, $\min \|x\|_1$ s.t. $Hx = b$, in two ways:

- **IRLS** (Daubechies–DeVore–Fornasier–Güntürk [2]). Repeat $x^{k+1} = \arg \min_{Hx=b} \sum_j w_j^k |x_j|^2$

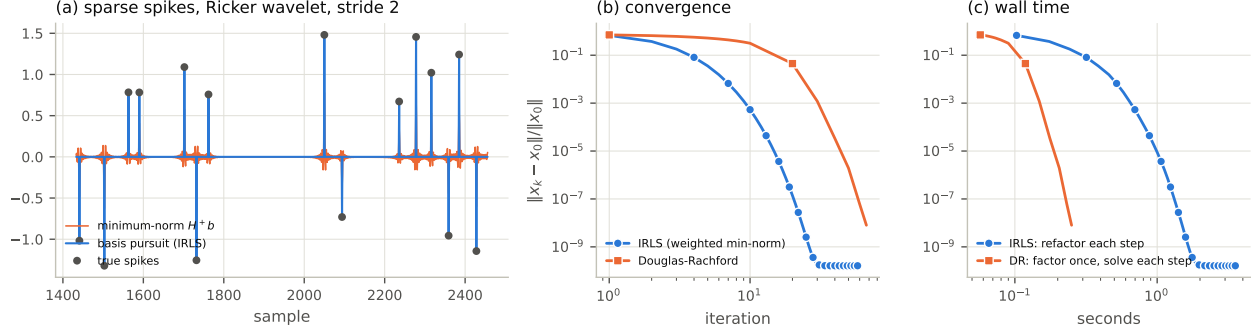


Figure 3: Basis pursuit for sparse-spike deconvolution. (a) Detail of the recovered spike train. (b, c) Convergence in iterations and in wall time.

with $w_j^k = (|x_j^k|^2 + \varepsilon_k^2)^{-1/2}$ and $\varepsilon_{k+1} = \min(\varepsilon_k, |x^{k+1}|_{(s+1)}/n)$. The inner problem is exactly (W) with $L = \text{diag}(\sqrt{w^k})$. The weights change every step, so each step is a fresh factorization of HL_k^{-1} , and Theorem 3.3 guarantees each inner solve is the exact weighted minimum-norm point.

- **Douglas–Rachford** on $\|x\|_1 + \iota_{\{Hx=b\}}$: $x^k = P(z^k)$, $z^{k+1} = z^k + \text{soft}_\gamma(2x^k - z^k) - x^k$, $\gamma = 0.05$, with the projection $P(v) = v + H^+(b - Hv)$. Here H never changes. It is factored *once* and each step is one $O(nr)$ solve plus two HSS products. ADMM for the same problem has the same inner step.

Results (Figure 3). The minimum-norm solution has relative error 0.71: it is smooth, not sparse. IRLS reaches 10^{-8} after 24 iterations and 1.6×10^{-10} after about 33; DR reaches 8×10^{-9} after 67. One factorization costs 0.056s and one solve 0.0023s, so a DR step is $\approx 24\times$ cheaper than an IRLS step, and DR wins in wall time (0.25s against 1.5s to 10^{-8}). The legacy MATLAB solver refactors on every call, so with it DR loses that advantage. In the current code H keeps its factors, so the projection $v + H(b - H*v)$ factors only on the first call.

6.2 Tikhonov deblurring and the choice of λ

The signal of M1, Sec. 7.3 ($n = 4096$), Gaussian blur $\sigma = 3$, stride 2, 1% white noise. The minimum-norm solution has error 94: it inverts the noise. A sweep of 23 values of λ costs 0.07s per λ (one factorization each). The discrepancy principle (largest λ with $\|Hx_\lambda - b\| \leq 1.01\|e\|$) gives $\lambda = 0.32$ with error 0.046, against the best on the grid, 0.036 at $\lambda = 0.56$ (Figure 4). The L-curve corner (maximum curvature) is at $\lambda = 0.032$, ten times smaller, with error 0.30: here the L-curve under-regularizes, a known limitation for smooth solutions [7]. The discrepancy principle needs $\|e\|$, which this test knows.

6.3 Non-Cartesian MRI with k -space weights

Problem (1-D model). Image x_0 on a centred grid of $N = 2048$ pixels (piecewise constant plus a bump). Data are $m = 921$ samples of its Fourier transform at variable-density non-Cartesian frequencies ξ_j , denser near $k = 0$. With $x'_j = -\xi_j \bmod 1$ this is exactly F5, with the Cartesian k -space grid l/N as the HSS column coordinates and $y = Fx$ as the unknown. A diagonal weight on y is therefore leaf-diagonal: the case this memo covers. The weight is $L = \text{diag}(1 + |k_l|/k_0)$, i.e. the Gaussian prior with power spectrum $(1 + |k|/k_0)^{-2}$ appropriate for piecewise-smooth images.

Results (Figure 5). Exact data: the image error is 0.120 for minimum norm and 0.098 for the weighted solution at $k_0 = 8$. The optimum over $k_0 \in \{1, \dots, 64\}$ is shallow, between 0.098 and 0.105.

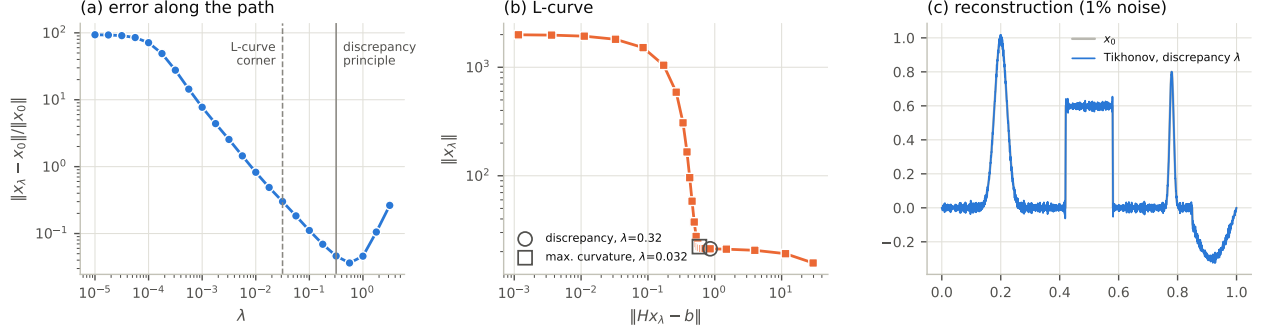


Figure 4: Tikhonov deblurring: error along the path, L-curve (circle: discrepancy principle, square: maximum curvature), and the discrepancy-principle reconstruction.

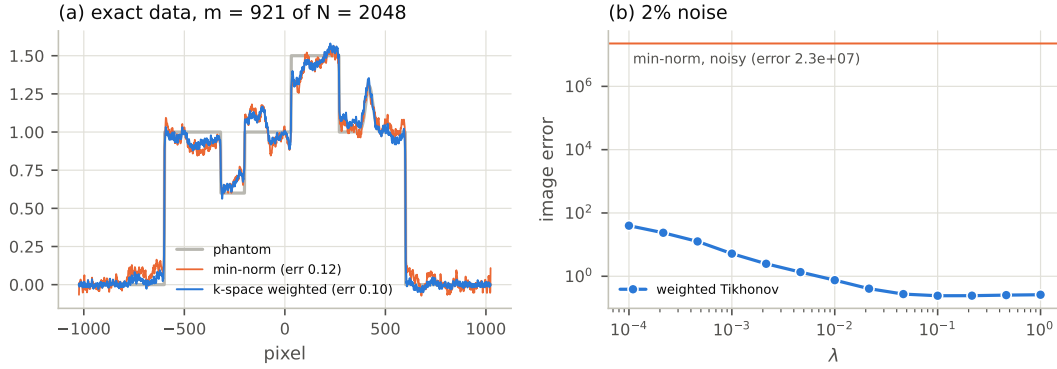


Figure 5: 1-D non-Cartesian MRI. (a) Exact data: minimum norm against k -space-weighted minimum norm. (b) 2% noise: weighted Tikhonov along the path against the unregularised minimum-norm image.

With 2% noise the unweighted minimum-norm image is useless (error 2×10^7): the variable-density nodes make C severely ill-conditioned. Weighted Tikhonov at $\lambda = 0.1$ restores an error of 0.24. *Tree*. Variable density is where the constructor’s index-based split hurts most: maximum HSS rank 135 with it, against 36 with the geometry-aligned tree (`mp_tree_aligned`).

6.4 Where the weights are not diagonal

Two items from the list need weights that this memo does not cover (Remark 3.5):

- *Seismic MWNI*: the HSS coordinates are the regular *spatial* grid, but the weights live on wavenumbers, so $L = F^* \text{diag}(P^{-1/2})F$ is circulant.
- *Image-domain sparsity in MRI* (compressed sensing): the HSS coordinates are k -space, but the IRLS weights live on pixels, which is again circulant.

Both are covered once L is allowed to be an HSS matrix (circulants with smooth symbols are). That needs HSS inversion and HSS products, which the repo does not have yet: the natural next step. Alternatively, ADMM can split the weight off into its own step; then only the unweighted minimum-norm solve is needed, in the HSS coordinates.

7 Open Items

1. **Factor/solve split:** done. H keeps its factors after the first solve, and `minnorm` and `tikhonov` keep theirs for the last weight and the last (λ, L, S) (M1, Sec. 8). Douglas–Rachford and ADMM need many solves with one matrix, and with stored factors a solve is $\approx 24\times$ cheaper than a factorization in the basis-pursuit test (Section 6.1) and $4\text{--}16\times$ at the sizes of M1, Table 5. A Tikhonov λ sweep still needs a new factorization per λ . A factorization that reuses the λ -independent parts (the U -compressions are the same for every λ) would cut that, and is worth working out.
2. **HSS weights.** Circulant or smoothing weights (MWNI, image-domain sparsity) need L itself HSS, and with it HSS inversion and products (Remark 3.5).
3. **General-form Tikhonov with a difference operator** (L a non-square $(n-1) \times n$ first difference) is not covered by the substitution. It needs the A -weighted pseudo-inverse (Eldén’s standard-form transformation) or a stacked formulation with an extra block row.
4. **Geometric tree** for variable-density sampling (rank 135 against 36 here; M1, Sec. 8).

A Code

MATLAB methods of `@hss`: `x = minnorm(H,b,'Weight',L)` and `[x,r] = tikhonov(H,b,lambda,'Weight',L,'DataWeight',S)`, where L and S are vectors (diagonal) or cell arrays of leaf blocks; omitted means identity. Both keep their factors with H (a repeat call with the same weights and λ only solves). Generator edits in `@hss/private/`: `hss_to_gen`, `hss_from_gen`, `hss_gen_weight` (leaf blocks applied by solves, never by `inv`), `hss_gen_augment`. The suite’s `mp_weighted_minnorm` and `mp_tikhonov` are thin wrappers. Tests: `mp_selftest`, `mp_test_solver` and `test_hss.m` (leaf-block weights against dense formulas), `mp_exp_weighted_tikhonov`, `mp_exp_apps2`. Python: `hssmn.variants` (`weighted_minnorm`, `tikhonov`, `augment`); experiments `exp_w.t.py`, `exp_apps2.py`; figures `fig_memo2.py`.

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